Assignment 3: Multivariate Gaussian Geometry and Relative Entropy

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1 Density Contours and Operator Characterization

Consider a multivariate normal distribution vector $X \sim \mathcal{N}(0, S)$ mapping an internal covariance architecture $S = A^T A$. The structural orientation of the probability density fields is governed by the quadratic formulation inside the exponential mapping function:

$$\Phi(x) = x^T S^{-1} x$$

The geometric boundary isolines where the density remains completely invariant represent ellipsoidal surfaces. By mapping the singular value decomposition features of A into the spectral characteristics of S:

- The orthogonal basis vectors v_i resulting from the diagonalization of S dictate the primary spatial directions of the ellipsoidal axes.
- The magnitude scales along these spatial directions are determined by the singular components σ_i, which are the square roots of the eigenvalues.

1.1 Singular Transformation Matrix Implication

When the mapping operator A is strictly singular, the determinant vanishes ($\det(A) = 0$), implying S cannot be inverted. Consequently, the standard probability density function collapses due to the non-existence of S^{-1}. The probability mass is constrained entirely within a lower-dimensional subspace, degenerating into a singular continuous measure.

2 Kullback-Leibler Divergence Formalism

2.1 Continuous Gaussian Systems

Evaluating the continuous cross-entropy functional between a baseline normal $\mathcal{N}(0, 1)$ and a shifted configuration $\mathcal{N}(\mu, \sigma^2)$ using expectation identities leads to the closed-form representation:

$$D_{\mathrm{KL}}(\mathcal{N} \parallel \tilde{\mathcal{N}}) = \frac{1}{2}\left[\ln(\sigma^2) + \frac{1+\mu^2}{\sigma^2} - 1\right]$$

2.2 Discrete Binary Bernoulli Spaces

For two independent Bernoulli models parameterized by p and q, the relative entropy is explicitly formulated as:

$$D_{\mathrm{KL}}(B_p \parallel B_q) = p \ln \left(\frac{p}{q}\right) + (1-p) \ln \left(\frac{1-p}{1-q}\right)$$

Substituting the constraint $q = 1 - p$ validates the directional identity symmetry $D_{\mathrm{KL}}(B_p \parallel B_{1-p}) = D_{\mathrm{KL}}(B_{1-p} \parallel B_p)$. Reversing the internal log ratios via inversion principles shows that the algebraic structure preserves its numerical value under this specific reflection mapping.